PERFORMANCE TESTING FOR EMPLOYEE DATA IMPORT S YSTEM

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TEAMID	**SWTID-2026-8059**
PROJECTTITLE	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
MAXIMUMMARKS	4Marks

1. Introduction

Performance testing is an important part of software development that evaluates how a system behaves under different workloads. In this project, performance testing is used to analyze how efficiently the ServiceNow system handles employee data import and incident assignment when multiple users access the system simultaneously.

The project focuses on importing employee data from external CSV files into ServiceNow using Import Sets and Transform Maps. When the number of records or users increases, the system should still maintain good response time and stability. Performance testing helps identify whether the system can handle large volumes of employee data and multiple incident requests without slowing down.

2. Objective of Performance Testing in the Project

The main objectives of performance testing in this project are:

To test the system performance when importing large employee datasets.

To measure the response time when users access employee records.

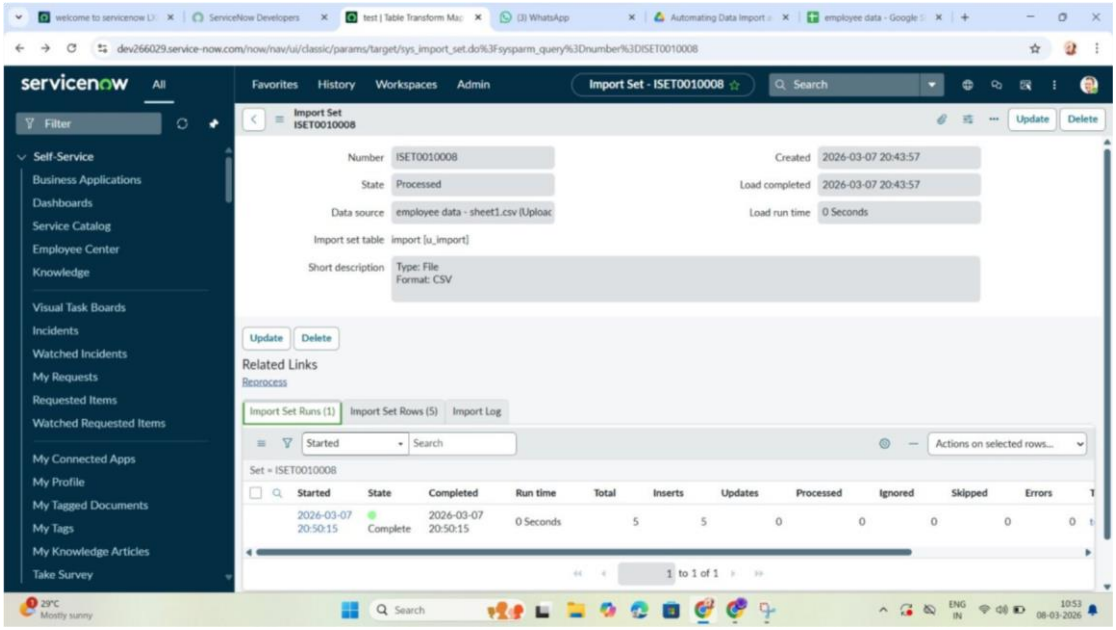
To evaluate system behavior when multiple incidents are created simultaneously.

To ensure that ServiceNow handles data transformation efficiently.

To identify performance bottlenecks during data import.

3. Performance Testing Scenario

In this project, performance testing is performed under the followingscenarios.



Scenario 1 – Employee Data Import

A large CSV file containing employee information is imported into ServiceNow. The system should process the data quickly without errors.

Scenario 2 – Multiple User Access

Multiple users access the system simultaneously to view employee records and create incidents.

Scenario 3 – Incident Assignment

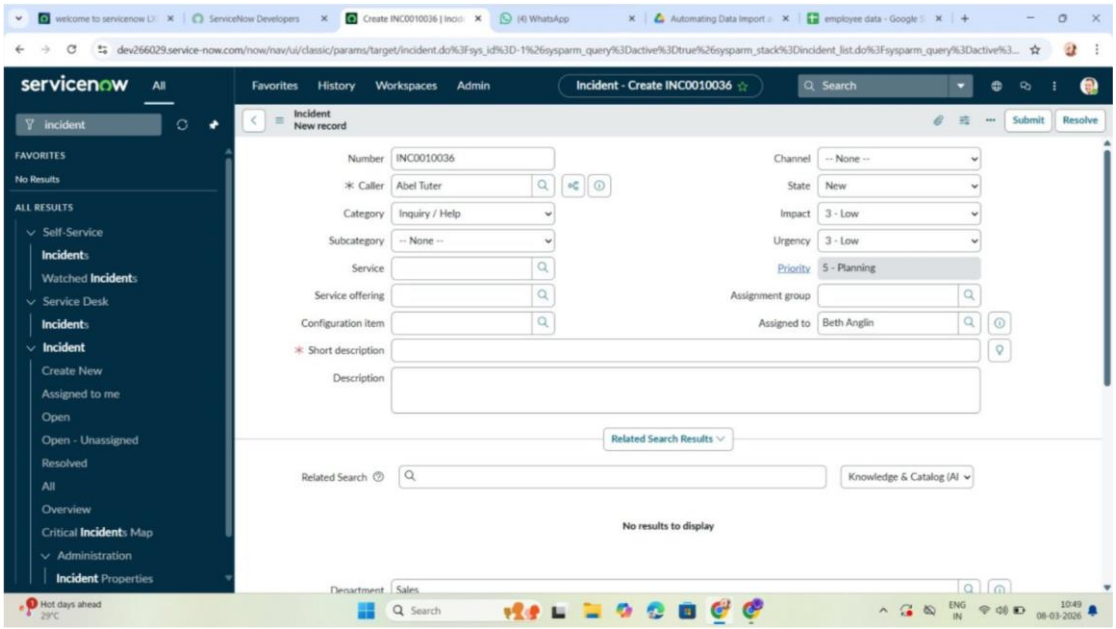
Service desk agents assign incidents to employees. The system should quickly retrieve department and manager details using dot walking.

Scenario 4 – Large Dataset Processing

The system is tested with thousands of employee records to check how well it handles large datasets.

4. Performance Testing Process

The performance testing process for this project includes several steps.



Step 1 – Test Planning

Define the performance testing objectives, expected number of users, and the size of the employee dataset.

Step 2 – Test Script Creation

Test scripts are created to simulate user actions such as importing data, viewing employee records, and creating incidents.

Step 3 – Test Environment Setup

A ServiceNow testing environment is configured to simulate real system conditions.

Step 4 – Test Execution

Multiple users are simulated to perform actions on the system simultaneously.

Step 5 – Monitoring

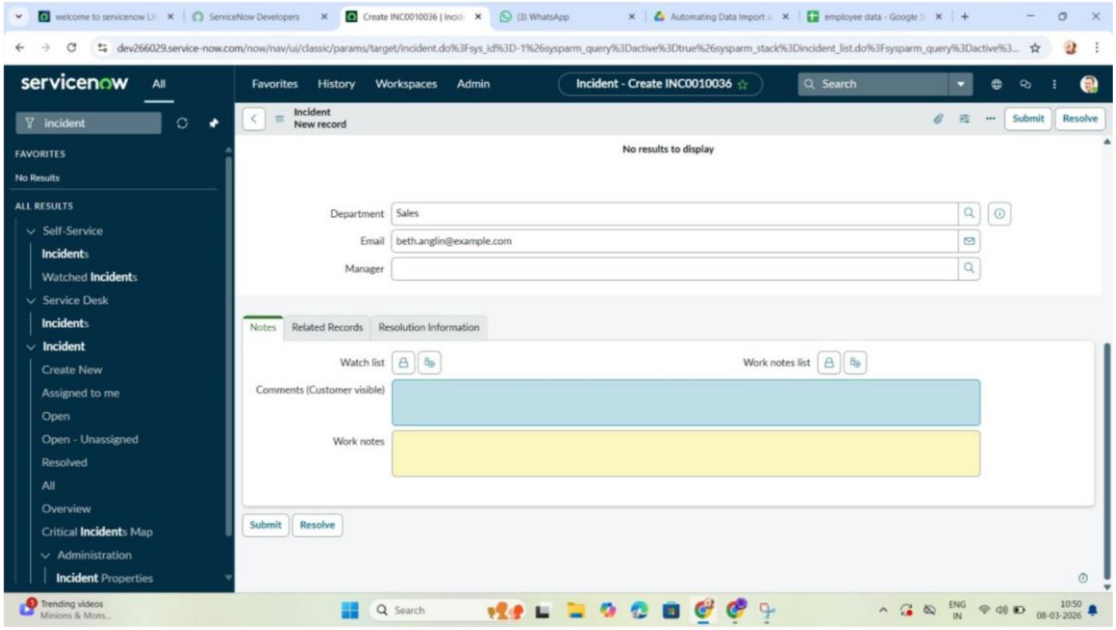
During testing, system performance metrics such as response time, CPU usage, and database performance are monitored.

Step 6 – Result Analysis

Test results are analyzed to identify slow operations or system bottlenecks.

5. Performance Metrics Used

The following metrics are used to evaluate system performance.



ResponseTime

Measures how quickly the system responds when a user requests employee information.

Throughput

Measures how many requests the system can process within a given time.

Data Processing Time

Measures how long it takes to import and transform employee records.

Error Rate

Indicates the number of failed requests during testing.

6. Expected Results

After conducting performance testing, the system should:

Import employee data efficiently.

Handle multiple users accessing records simultaneously.

Display related employee information quickly.

Maintain stable performance under heavy workloads.

7. Conclusion

Performance testing ensures that the employee data import system functions efficiently even when large datasets are processed or when many users access the system simultaneously. By analyzing system performance and optimizing the process, the system becomes more reliable and scalable.

Project Related Performance Testing Diagram

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|   Admin / HR System |
| Upload Employee CSV |
+----------+-----------+
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           v
+----------------------+
|   ServiceNow Import |
|     Set Table       |
+----------+-----------+
           |
           v
+----------------------+
|    Transform Map     |
|   Data Processing    |
+----------+-----------+
           |
           v
+----------------------+
|   ServiceNow User    |
|    Database          |
+----------+-----------+
```

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          |
          v
+----------------------+
| Multiple Users       |
| Access Employee Data |
+----------+-----------+
          |
          v
+----------------------+
| Incident Creation &  |
| Dot Walking Fetching |
+----------+-----------+
          |
          v
+----------------------+
| Performance Metrics  |
| Response Time / Load |
+----------------------+
```